Problem Set 5: Cellular Respiration

Due: Sunday, November 1 at 11:59 PM **Value: 25 points** **Format: Five-question problem set**

Purpose

Trace matter, electrons, proton gradients, and ATP production through cellular respiration.

Your task

- Q1 (5): Account for all six carbons of glucose from glycolysis through the citric acid cycle.
- Q2 (5): Explain why NADH production can continue briefly after ATP synthase is inhibited but eventually slows.
- Q3 (5): Predict oxygen consumption, proton gradient, and ATP production after adding an uncoupler.
- Q4 (5): Compare substrate-level phosphorylation with oxidative phosphorylation by mechanism and location.
- Q5 (5): Use the respirometer data to calculate corrected oxygen consumption for germinating and nongerminating seeds.

Provided evidence or data

Chamber	Initial reading	Final reading	Temp-control change
Germinating seeds	0.12 mL	0.49 mL	0.04 mL
Nongerminating seeds	0.10 mL	0.19 mL	0.04 mL

Submit

- One PDF with a labeled pathway diagram and calculations.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.